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Carlson Nitronic® 60® Austenitic Stainless Steel

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Material General Description

Notes:

Nitronic 60 is an anti-galling and wear resistant austenitic stainless steel. It provides a significantly lower cost alternative to combat wear and galling when compared with cobalt-bearing and high nickel alloys. The uniform corrosion resistance of Nitronic 60 is superior to 304 in most media. Chloride pitting resistance is superior to 316. Room temperature Nitronic 60 also possesses excellent high temperature oxidation resistance and low temperature impact resistance.

Due to its grain structure, the galling resistance of Nitronic 60 is superior to that of any other stainless steel. It resists wear, tearing or freeze up of both the primary and contact materials over a wide range of service temperatures. The elevated temperature wear resistance of Nitronic 60 is excellent despite its relatively low hardness when compared with cobalt and nickel-base wear alloys. It also performs well in metal-to-metal wear in nominally inert atmospheres. The cavitation erosion resistance of Nitronic 60 is superior to the austenitic stainless steels as well as high strength duplex (ferritic-austenitic) stainless steels.

The general corrosion resistance of Nitronic 60 falls between 304 and 316. However, in a wear system, a galling or seizure failure occurs first followed by dimensional loss due to wear and finally corrosion. Although the general corrosion resistance of Nitronic 60 is not as good as 316, it does offer better chloride pitting, stress corrosion cracking and crevice corrosion resistance.

In high temperature service, Nitronic 60 exhibits carburization superior to 316L and 309. Its oxidation resistance is far superior to 304 and 316 and comparable to 309.

Applications

- Infrastructure bridge pin and hanger expansion joints, parking deck expansion joint wear plates.
- Hydroelectric Power – stems, wicket gate wear rings
- Oil and Gas Production – pump wear rings, bushings, valve trim, seals, fittings, logging equipment and screens.
- Food processing and Pharmaceuticals – galling resistant applications in sanitary equipment where lubricants can not be used
- Chemical and Petrochemical – process valve stems, seats and trim, pump wear rings.

Information provided by Carlson

Key Words: UNS S21800

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Physical Properties	Original Value	Comments
Density	7.62 g/cc	
Mechanical Properties	Original Value	Comments
Hardness, Brinell	<= 241	
Hardness, Rockwell B	<= 100	
Tensile Strength at Break	91500 psi	Carburized
	>= 95000 psi	
Tensile Strength, Yield	>= 50000 psi	
	@Strain 0.2 %	
	58000 psi	Carburized
	@Strain 0.2 %	
Elongation at Break	19 %	Carburized; in 4xD
	>= 35 %	
Reduction of Area	21.5 %	Carburized
Modulus of Elasticity	26200 ksi	
Taber Abrasion, mg/1000 Cycles	3.2	vs. Nitronic 32
	6.0	vs. 304
Bend Radius, Minimum	1.5 t	Carburized; 100°
Electrical Properties	Original Value	Comments
Electrical Resistivity	0.0000982 ohm-cm	
Thermal Properties	Original Value	Comments
CTE, linear	8.80 $\mu\text{in/in-}^{\circ}\text{F}$	
	@Temperature 25.6 - 93.3 $^{\circ}\text{C}$	
	9.60 $\mu\text{in/in-}^{\circ}\text{F}$	
	@Temperature 25.6 - 316 $^{\circ}\text{C}$	
	10.3 $\mu\text{in/in-}^{\circ}\text{F}$	
	@Temperature 25.6 - 649 $^{\circ}\text{C}$	
	11.0 $\mu\text{in/in-}^{\circ}\text{F}$	
	@Temperature 25.6 - 982 $^{\circ}\text{C}$	
Component Elements Properties	Original Value	Comments
Carbon, C	<= 0.10 %	
Chromium, Cr	16 - 18 %	
Iron, Fe	59.13 - 65.42 %	
Manganese, Mn	7.0 - 9.0 %	
Nickel, Ni	8.0 - 9.0 %	
Nitrogen, N	0.080 - 0.18 %	
Phosphorus, P	<= 0.060 %	
Silicon, Si	3.5 - 4.5 %	
Sulfur, S	<= 0.030 %	

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